

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12412048
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Sang; Yi Shan et al.

Automatically integrating user translation feedback

Abstract

A method includes: receiving, by a computing device, user input indicating an incorrect translation of a word appearing in an interface of an application; identifying, by the computing device, other instances of the word in other interfaces of the application, wherein the identifying is performed using a glossary relationship set that is based on association analysis, wherein the other instances of the word constitute less than all instances of the word appearing in all interfaces of the application; and generating, by the computing device, a new version of the application having a revised translation of the word in the interface and the other instances of the word in the other interfaces.

Inventors: Sang; Yi Shan (Xian, CN), Yang; Jing (Xian, CN), Liu; Chong (Xian, CN), Xu; Jin (Xian, CN), Li; Chao (Xian, CN), Lu; Xu (Xian, CN)

Applicant: INTERNATIONAL BUSINESS MACHINES CORPORATION (Armonk, NY)

Family ID: 86768420

Assignee: International Business Machines Corporation (Armonk, NY)

Appl. No.: 17/557737

Filed: December 21, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230196034 A1	Jun. 22, 2023

Publication Classification

Int. Cl.: G06F40/51 (20200101); G06F8/658 (20180101); G06F40/242 (20200101)

U.S. Cl.:

CPC **G06F40/51** (20200101); **G06F8/658** (20180201); **G06F40/242** (20200101);

Field of Classification Search

CPC: G06F (40/51); G06F (8/658); G06F (40/242); G06F (40/47); G06F (9/454)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6401091	12/2001	Butler	707/999.005	G06F 16/907
6937974	12/2004	d'Agostini	704/277	G06F 40/44
9342499	12/2015	Madnani et al.	N/A	N/A
9552354	12/2016	Seligman	N/A	G06F 40/51
10346543	12/2018	Leydon et al.	N/A	N/A
10650086	12/2019	Knudson	N/A	G06F 40/134
11003426	12/2020	Wilson	N/A	G06F 9/45508
2002/0169803	12/2001	Sampath	715/234	G06F 40/106
2003/0101044	12/2002	Krasnov	704/4	G06F 40/55
2003/0189603	12/2002	Goyal	704/E15.04	G10L 15/22
2005/0075858	12/2004	Pournasseh	704/2	G06F 40/58
2006/0047689	12/2005	Gabbert	707/999.102	G06Q 10/10
2008/0195377	12/2007	Kato	704/8	G06F 40/47
2009/0299732	12/2008	Hao	704/10	G06F 40/289
2011/0202330	12/2010	Dai	704/9	G06F 40/268
2012/0166984	12/2011	Brunswig	715/765	G06F 3/04847
2012/0323557	12/2011	Koll	704/8	G10L 15/183
2013/0035927	12/2012	Cha	704/3	G06F 40/58
2013/0124987	12/2012	Lakritz	715/264	H04L 63/0281
2013/0300636	12/2012	Cunningham	345/8	G06F 3/013

2014/0081620	12/2013	Solntseva	704/3	G06F 3/04883
2015/0032712	12/2014	Green	707/706	G06F 16/285
2015/0120273	12/2014	Gusakov	704/2	G06F 40/58
2015/0363394	12/2014	Marciano et al.	N/A	N/A
2016/0224654	12/2015	Tsuchida	N/A	G06N 20/00
2016/0232156	12/2015	Orsini et al.	N/A	N/A
2017/0235722	12/2016	Leydon	704/2	G06F 40/51
2018/0123814	12/2017	Sexauer	N/A	H04L 51/04
2018/0129634	12/2017	Sivaji	N/A	G06F 40/106
2018/0225278	12/2017	Alba	N/A	G06F 40/30
2019/0171792	12/2018	Manica	N/A	G06F 40/284
2020/0012717	12/2019	Brown	N/A	G06F 40/205
2020/0160458	12/2019	Bodin	N/A	H04L 67/53
2021/0019373	12/2020	Freitag	N/A	G06F 40/58
2021/0216727	12/2020	Liu	N/A	G06F 40/47
2021/0264804	12/2020	Venkatasubramanyam	N/A	G06N 5/04
2022/0294748	12/2021	Crowley	N/A	G06F 40/35

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
108563643	12/2017	CN	N/A

OTHER PUBLICATIONS

Disclosed Anonymously, “A method of self-improvement translation with machine translation and social translation”, IP. com, IPCOM000244449D, Dec. 13, 2015, 6 pages. cited by applicant
Rozovskaya et al., “Grammatical Error Correction: Machine Translation and Classifiers”,
<https://www.researchgate.net/publication/306093686>, Proceedings of the 54th Annual Meeting of
the Association for Computational Linguistic, Aug. 12, 2016, 12 pages. cited by applicant

Primary Examiner: Mehta; Bhavesh M

Assistant Examiner: Lam; Philip H

Attorney, Agent or Firm: Calderon Safran & Wright P.C.

Background/Summary

BACKGROUND

(1) Aspects of the present invention relate generally to translation systems and, more particularly, to adjusting translations in software applications based on user feedback.

(2) Software applications that generate user interfaces, including dialog boxes, are sometimes translated from a first language to a second language. The applications sometimes include polysemous words in the user interfaces.

SUMMARY

(3) In a first aspect of the invention, there is a computer-implemented method including: receiving, by a computing device, user input indicating an incorrect translation of a word appearing in an interface of an application; identifying, by the computing device, other instances of the word in other interfaces of the application, wherein the identifying is performed using a glossary relationship set that is based on association analysis, wherein the other instances of the word constitute less than all instances of the word appearing in all interfaces of the application; and generating, by the computing device, a new version of the application having a revised translation of the word in the interface and the other instances of the word in the other interfaces.

(4) In another aspect of the invention, there is a computer program product including one or more computer readable storage media having program instructions collectively stored on the one or more computer readable storage media. The program instructions are executable to: receive user input indicating an incorrect translation of a word appearing in an interface of an application; identify other instances of the word in other interfaces of the application, wherein the identifying is performed using a glossary relationship set that is based on association analysis, wherein the other instances of the word constitute less than all instances of the word appearing in all interfaces of the application; and generate a new version of the application having a revised translation of the word in the interface and the other instances of the word in the other interfaces.

(5) In another aspect of the invention, there is system including a processor, a computer readable memory, one or more computer readable storage media, and program instructions collectively stored on the one or more computer readable storage media. The program instructions are executable to: receive user input indicating an incorrect translation of a word appearing in an interface of an application; identify other instances of the word in other interfaces of the application, wherein the identifying is performed using a glossary relationship set that is based on association analysis, wherein the other instances of the word constitute less than all instances of the word appearing in all interfaces of the application; and generate a new version of the application having a revised translation of the word in the interface and the other instances of the word in the other interfaces.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present invention are described in the detailed description which follows, in reference to the noted plurality of drawings by way of non-limiting examples of exemplary embodiments of the present invention.

(2) FIG. 1 depicts a computer infrastructure in accordance with aspects of the invention.

(3) FIG. 2 shows a block diagram of an exemplary environment in accordance with aspects of the invention.

(4) FIGS. 3A and 3B show examples of dialog boxes and dialogs in accordance with aspects of the invention.

- (5) FIG. 4 shows a block diagram of an exemplary process of updating a translation of a word in an application in accordance with aspects of the invention.
- (6) FIG. 5 shows a block diagram of an exemplary process of generating glossary relationships in accordance with aspects of the invention.
- (7) FIG. 6 shows a flowchart of an exemplary method of generating glossary relationships in accordance with aspects of the invention.
- (8) FIG. 7 shows an example of determining glossary relationships for a category in accordance with aspects of the invention.
- (9) FIG. 8 shows exemplary mappings of dialogs to glossary relationships in accordance with aspects of the invention.
- (10) FIG. 9 shows a flowchart of an exemplary method in accordance with aspects of the invention.

DETAILED DESCRIPTION

(11) Aspects of the present invention relate generally to translation systems and, more particularly, to adjusting translations in software applications based on user feedback. Manual translation and machine translation of software from a first language to a second language often results in incorrectly translated words. Because the user interfaces of applications typically have mostly words and short phrases, it is nearly impossible to determine context directly from a resource file, with the result being that it is very difficult to accurately translate the words that appear in the user interfaces. When a translated word shown in the user interface is incorrect and needs updated, it is difficult to determine which other instances of this same word should also have their translation changed. This is due to some words having a same spelling but different meanings. For example, the polysemous word “log” may refer to a written record of events, a mathematical function, a part of a tree, or an act of entering data. As a result of these different meanings, an incorrect translation of the word “log” in one user interface does not necessarily mean that it is appropriate to change the same word in other instances in other user interfaces of the same application. Moreover, when a problem is found in the translation of one language, it is not easy to judge whether the same translation problem exists in other languages.

(12) Aspects of the invention address this problem by providing a system and method that use clustering and association analysis to determine which instances of a translated word to change, and which not to change, in response to receiving an indication of an updated translation from a user of a software application. Implementations of the invention group resource words (or phrases) based upon functional use cases. Embodiments use cluster analysis on the resource groups and association relationship analysis on a glossary subset to obtain a glossary association relationship set. In response to receiving user specific feedback to auto update a translated word (or phrase), the system uses the glossary association relationship set to determine which instances of the word to update in the resource file (and which instances of the same word to not update). In this manner, implementations of the invention provide for automatically clustering the word resource group to the same functional category and using association relationship analysis to obtain a glossary association relationship set. Implementations also provide for automatically verifying user feedback and updating the word's translation in the glossary association relationship set, thereby providing complete multi-language mutual translation verification. Implementations also provide for automatically selecting all resource groups that are related via the glossary association relationship set and updating these translations at the same time. Implementations also provide for automatically integrating the updated translation into a product system. Implementations also provide for automatically aggregating words of a same functional category and using the association relationship to determine usage scenarios of the words and sentences in the glossary dictionary, thereby improving the accuracy of translation. Implementations also provide for automatically verifying user feedback and complete multi-language mutual translation verification. Implementations also provide for automatically determining all related resource groupings according to the association relationship of glossary and updating all the same translation

questions. Implementations also provide for automatically integrating product development, quickly responding to customer feedback, and improving customer satisfaction.

(13) The present invention may be a system, a method, and/or a computer program product at any possible technical detail level of integration. The computer program product may include a computer readable storage medium (or media) having computer readable program instructions thereon for causing a processor to carry out aspects of the present invention.

(14) The computer readable storage medium can be a tangible device that can retain and store instructions for use by an instruction execution device. The computer readable storage medium may be, for example, but is not limited to, an electronic storage device, a magnetic storage device, an optical storage device, an electromagnetic storage device, a semiconductor storage device, or any suitable combination of the foregoing. A non-exhaustive list of more specific examples of the computer readable storage medium includes the following: a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), a static random access memory (SRAM), a portable compact disc read-only memory (CD-ROM), a digital versatile disk (DVD), a memory stick, a floppy disk, a mechanically encoded device such as punch-cards or raised structures in a groove having instructions recorded thereon, and any suitable combination of the foregoing. A computer readable storage medium or media, as used herein, is not to be construed as being transitory signals per se, such as radio waves or other freely propagating electromagnetic waves, electromagnetic waves propagating through a waveguide or other transmission media (e.g., light pulses passing through a fiber-optic cable), or electrical signals transmitted through a wire.

(15) Computer readable program instructions described herein can be downloaded to respective computing/processing devices from a computer readable storage medium or to an external computer or external storage device via a network, for example, the Internet, a local area network, a wide area network and/or a wireless network. The network may comprise copper transmission cables, optical transmission fibers, wireless transmission, routers, firewalls, switches, gateway computers and/or edge servers. A network adapter card or network interface in each computing/processing device receives computer readable program instructions from the network and forwards the computer readable program instructions for storage in a computer readable storage medium within the respective computing/processing device.

(16) Computer readable program instructions for carrying out operations of the present invention may be assembler instructions, instruction-set-architecture (ISA) instructions, machine instructions, machine dependent instructions, microcode, firmware instructions, state-setting data, configuration data for integrated circuitry, or either source code or object code written in any combination of one or more programming languages, including an object oriented programming language such as Smalltalk, C++, or the like, and procedural programming languages, such as the "C" programming language or similar programming languages. The computer readable program instructions may execute entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider). In some embodiments, electronic circuitry including, for example, programmable logic circuitry, field-programmable gate arrays (FPGA), or programmable logic arrays (PLA) may execute the computer readable program instructions by utilizing state information of the computer readable program instructions to personalize the electronic circuitry, in order to perform aspects of the present invention.

(17) Aspects of the present invention are described herein with reference to flowchart illustrations and/or block diagrams of methods, apparatus (systems), and computer program products according to embodiments of the invention. It will be understood that each block of the flowchart illustrations

and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer readable program instructions.

(18) These computer readable program instructions may be provided to a processor of a computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks. These computer readable program instructions may also be stored in a computer readable storage medium that can direct a computer, a programmable data processing apparatus, and/or other devices to function in a particular manner, such that the computer readable storage medium having instructions stored therein comprises an article of manufacture including instructions which implement aspects of the function/act specified in the flowchart and/or block diagram block or blocks.

(19) The computer readable program instructions may also be loaded onto a computer, other programmable data processing apparatus, or other device to cause a series of operational steps to be performed on the computer, other programmable apparatus or other device to produce a computer implemented process, such that the instructions which execute on the computer, other programmable apparatus, or other device implement the functions/acts specified in the flowchart and/or block diagram block or blocks.

(20) The flowchart and block diagrams in the Figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods, and computer program products according to various embodiments of the present invention. In this regard, each block in the flowchart or block diagrams may represent a module, segment, or portion of instructions, which comprises one or more executable instructions for implementing the specified logical function(s). In some alternative implementations, the functions noted in the blocks may occur out of the order noted in the Figures. For example, two blocks shown in succession may, in fact, be accomplished as one step, executed concurrently, substantially concurrently, in a partially or wholly temporally overlapping manner, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustration, and combinations of blocks in the block diagrams and/or flowchart illustration, can be implemented by special purpose hardware-based systems that perform the specified functions or acts or carry out combinations of special purpose hardware and computer instructions.

(21) Referring now to FIG. 1, a schematic of an example of a computer infrastructure is shown. Computer infrastructure **10** is only one example of a suitable computer infrastructure and is not intended to suggest any limitation as to the scope of use or functionality of embodiments of the invention described herein. Regardless, computer infrastructure **10** is capable of being implemented and/or performing any of the functionality set forth hereinabove.

(22) In computer infrastructure **10** there is a computer system **12**, which is operational with numerous other general purpose or special purpose computing system environments or configurations. Examples of well-known computing systems, environments, and/or configurations that may be suitable for use with computer system **12** include, but are not limited to, personal computer systems, server computer systems, thin clients, thick clients, hand-held or laptop devices, multiprocessor systems, microprocessor-based systems, set top boxes, programmable consumer electronics, network PCs, minicomputer systems, mainframe computer systems, and distributed cloud computing environments that include any of the above systems or devices, and the like.

(23) Computer system **12** may be described in the general context of computer system executable instructions, such as program modules, being executed by a computer system. Generally, program modules may include routines, programs, objects, components, logic, data structures, and so on that perform particular tasks or implement particular abstract data types. Computer system **12** may be practiced in distributed cloud computing environments where tasks are performed by remote

processing devices that are linked through a communications network. In a distributed cloud computing environment, program modules may be located in both local and remote computer system storage media including memory storage devices.

(24) As shown in FIG. 1, computer system **12** in computer infrastructure **10** is shown in the form of a general-purpose computing device. The components of computer system **12** may include, but are not limited to, one or more processors or processing units **16**, a system memory **28**, and a bus **18** that couples various system components including system memory **28** to processor **16**.

(25) Bus **18** represents one or more of any of several types of bus structures, including a memory bus or memory controller, a peripheral bus, an accelerated graphics port, and a processor or local bus using any of a variety of bus architectures. By way of example, and not limitation, such architectures include Industry Standard Architecture (ISA) bus, Micro Channel Architecture (MCA) bus, Enhanced ISA (EISA) bus, Video Electronics Standards Association (VESA) local bus, and Peripheral Component Interconnects (PCI) bus.

(26) Computer system **12** typically includes a variety of computer system readable media. Such media may be any available media that is accessible by computer system **12**, and it includes both volatile and non-volatile media, removable and non-removable media.

(27) System memory **28** can include computer system readable media in the form of volatile memory, such as random access memory (RAM) **30** and/or cache memory **32**. Computer system **12** may further include other removable/non-removable, volatile/non-volatile computer system storage media. By way of example only, storage system **34** can be provided for reading from and writing to a non-removable, non-volatile magnetic media (not shown and typically called a “hard drive”). Although not shown, a magnetic disk drive for reading from and writing to a removable, non-volatile magnetic disk (e.g., a “floppy disk”), and an optical disk drive for reading from or writing to a removable, non-volatile optical disk such as a CD-ROM, DVD-ROM or other optical media can be provided. In such instances, each can be connected to bus **18** by one or more data media interfaces. As will be further depicted and described below, memory **28** may include at least one program product having a set (e.g., at least one) of program modules that are configured to carry out the functions of embodiments of the invention.

(28) Program/utility **40**, having a set (at least one) of program modules **42**, may be stored in memory **28** by way of example, and not limitation, as well as an operating system, one or more application programs, other program modules, and program data. Each of the operating system, one or more application programs, other program modules, and program data or some combination thereof, may include an implementation of a networking environment. Program modules **42** generally carry out the functions and/or methodologies of embodiments of the invention as described herein.

(29) Computer system **12** may also communicate with one or more external devices **14** such as a keyboard, a pointing device, a display **24**, etc.; one or more devices that enable a user to interact with computer system **12**; and/or any devices (e.g., network card, modem, etc.) that enable computer system **12** to communicate with one or more other computing devices. Such communication can occur via Input/Output (I/O) interfaces **22**. Still yet, computer system **12** can communicate with one or more networks such as a local area network (LAN), a general wide area network (WAN), and/or a public network (e.g., the Internet) via network adapter **20**. As depicted, network adapter **20** communicates with the other components of computer system **12** via bus **18**. It should be understood that although not shown, other hardware and/or software components could be used in conjunction with computer system **12**. Examples, include, but are not limited to: microcode, device drivers, redundant processing units, external disk drive arrays, RAID systems, tape drives, and data archival storage systems, etc.

(30) FIG. 2 shows a block diagram of an exemplary environment in accordance with aspects of the invention. In embodiments, the environment includes a network **201** that provides communication between a user device **205** and a translation server **210**. The network **201** includes one or more

communications networks such as one or more of a LAN, WAN, and the Internet. The user device **205** is a user computing device such as a desktop computer, laptop computer, tablet computer, smartphone, smartwatch, etc., and may comprise one or more elements of computer system **12** (FIG. **1**). In one example, the translation server **210** is one or more computing devices each including one or more elements of the computer system **12** of FIG. **1**. In another example, the translation server **210** is one or more virtual machines (VMs) or containers running on one or more computing devices.

(31) In embodiments, the user device **205** comprises an application **215** (e.g., a software application), which may comprise one or more programs such as program/utility **40** of FIG. **1**. In embodiments, the user device **205** also comprises a translation feedback module **220**, which may comprise one or more program modules such as program modules **42** of FIG. **1**. The application **215** may be a stand-alone application running locally on the user device **205** or may be a client application that communicates with a host application running remotely from the user device **205**. The translation feedback module **220** may be an extension or add-on to the application **215**. In embodiments, the application **215** causes the user device **205** to display a user interface (UI) **223** that includes words that have been translated from a product resource file that is written in another language. For example, the UI **223** may be configured to display plural different dialog boxes (e.g., interfaces) each containing plural English language words that have been translated from a product resource file that is written in Chinese. The application **215** can be any type of application such as, for example, a statistical analysis application that a user utilizes to perform statistical analysis of data.

(32) According to aspects of the invention, the translation feedback module **220** causes the UI **223** to display a translation feedback interface by which a user can provide input to indicate an issue (e.g., problem) with a translation of a word included in an interface of the application **215**. For example, a user may notice that one of the English language words in one of the dialog boxes does not appear to have been translated correctly, and the user may provide input in the translation feedback interface indicating this word. In accordance with aspects of the invention, the translation server **210** is configured to receive the user input indicating an incorrectly translated word in a particular dialog box, identify other instances of the same word in other dialog boxes based on relationships defined by a glossary relationship set, and provide a new version of the application **215** with a revised translation of the identified instances of the word. In accordance with aspects of the invention, the translation server **210** is configured to only revise the translation of other instances of the same word that have the relationship defined by the glossary relationship set, and not revise the translation of other instances of the same word that do not have the relationship defined by the glossary relationship set. In this manner, implementations of the invention revise the translation of some instances of a word in the application **215** but do not revise the translation of other instances of the same word in the application **215**, based on relationships defined by a glossary relationship set as described herein.

(33) In embodiments, the translation server **210** comprises a glossary relationship set generation module **225**, glossary relationship set application module **230**, translation engine module **235**, modification and compiler module **240**, and verification module **245**, each of which may comprise one or more program modules such as program modules **42** described with respect to FIG. **1**. The translation server **210** may include additional or fewer modules than those shown in FIG. **2**. In embodiments, separate modules may be integrated into a single module. Additionally, or alternatively, a single module may be implemented as multiple modules. Moreover, the quantity of devices and/or networks in the environment is not limited to what is shown in FIG. **2**. In practice, the environment may include additional devices and/or networks; fewer devices and/or networks; different devices and/or networks; or differently arranged devices and/or networks than illustrated in FIG. **2**.

(34) In accordance with aspects of the invention, the glossary relationship set generation module

225 is configured to generate a glossary relationship set as described herein. In embodiments, the glossary relationship set is a set of plural glossary relationships, where each of the glossary relationships defines a group of associated words from one of the dialog boxes of the application **215**. In implementations, the glossary relationship set generation module **225** determines the groups of associated words using clustering techniques (e.g., cosine similarity) and association analysis techniques (e.g., market basket analysis). In embodiments, the glossary relationship set generation module **225** maps the dialogs to the glossary relationships.

(35) In accordance with aspects of the invention, the glossary relationship set application module **230** is configured to identify other instances of a word in other dialog boxes of the application **215** based on the glossary relationship set. In embodiments, in response to receiving an indication of a word and a particular dialog box from the user, the glossary relationship set application module **230** determines a glossary relationship that is mapped to a dialog containing this word, and determines other dialogs that are mapped to the same glossary relationship and that have the same word. In this manner, the glossary relationship set application module **230** determines other instances of the same word that are related in a manner defined by the glossary relationship set.

(36) In accordance with aspects of the invention, the translation engine module **235** is configured to revise a translation of the word in portions of a product resource file that correspond to the words identified by the glossary relationship set application module **230**.

(37) In accordance with aspects of the invention, the modification and compiler module **240** is configured to create a new version of the application **215** using the output of the translation engine module **235**. In embodiments, the modification and compiler module **240** compiles, assembles, and integrates the modified product resource file to create a new version of the application **215**.

(38) In accordance with aspects of the invention, the verification module **245** is configured to verify the modified product resource file. In embodiments, the verification module **245** runs a translation verification test using a translation verification case that is based on the output of the translation engine module **235**.

(39) FIGS. **3A** and **3B** show examples of dialog boxes **311**, **312** displayed by the UI **223** of an exemplary statistical analysis type of application **215**. There may be any number of such dialog boxes in the application **215** including tens or even hundreds of different dialog boxes. Each dialog box may be associated with a function of the application **215**. For example, dialog box **311** may be associated with the function titled Power Analysis One-Sample Mean, and dialog box **312** may be associated with the function titled Bivariate Correlations. Dialog **D1** shown in FIG. **3A** is a list of all the words included in the dialog box **311** with high frequency words filtered (e.g., omitted). Dialog **D2** shown in FIG. **3B** is a list of all the words included in the dialog box **312** with high frequency words filtered (e.g., omitted). In embodiments, the glossary relationship set generation module **225** determines a respective dialog (e.g., **D1**, **D2**, etc.) for each dialog box of the application **215** by analyzing sections of the product resource file that correspond to the different dialog boxes. In this manner, for a product application that has “n” number of dialog boxes, the glossary relationship set generation module **225** determines **D1**, **D2**, . . . , **Dn** different dialogs. As described herein, the glossary relationship set generation module **225** uses the dialogs **D1-Dn** to determine a glossary relationship set for this application **215**. As further described herein, the glossary relationship set application module **230** uses the determined glossary relationship set to determine which instances of a word in the product resource file to change the translation of in response to receiving an indication from a user that the translation of the word is incorrect in one of the dialog boxes.

(40) FIG. **4** shows a functional block diagram of an exemplary process of updating a translation of a word in an application in accordance with aspects of the invention. Steps of the process may be carried out in the environment of FIG. **2** and are described with reference to elements depicted in FIG. **2**. At step **401**, a user **400** of the application **215** provides input via the translation feedback interface of the UI **223**, the input defining an issue (e.g., problem) with a translation of one of the

words in one of the dialog boxes of the application **215**. The input may comprise the English-language string (e.g., word), an original translation of the string, and an updated translation of the string. At step **402**, the server **210** uses a predefined glossary relationship set to determine a glossary relationship that contains the string (e.g., word) indicated by the user at step **401**. At step **403**, the server **210** identifies other instances of the same string (e.g., word) that occur in other dialog boxes, the identifying being based on the determined glossary relationship. At step **404**, the server **210** updates a force glossary, which is a document that forces certain terms (contained in the document) to be translated in a specific way (defined in the document). In embodiments, the server **210** updates the force glossary to translate the string (e.g., word) indicated by the user in a specific way indicated by the user. At step **405**, the server **210** modifies the product resource file of the application **215** by changing the translation of all instances of the word identified at step **403** using the force glossary definition that was updated at step **404**. At step **406**, the sever **210** generates a pull request (PR) to create a new version of the application **215** based on the modified product resource file from step **405**. At step **407**, the server **210** merges the modified product resource file with a code base. At step **408**, the sever **210** builds a new version of the application **215** using the modified product resource file, e.g., by compiling, assembling, and integrating a new executable version of the application **215**. At step **409**, the server **210** performs a translation verification test (TVT) of the modified product resource file using a translation verification case. At step **410**, the server **210** provides the new version of the application **215** to the user, e.g., via the user device **205**.

(41) FIG. 5 shows a functional block diagram of an exemplary process of generating glossary relationships in accordance with aspects of the invention. Steps of the process may be carried out in the environment of FIG. 2 and are described with reference to elements depicted in FIG. 2-4.

Elements **501a**, **501b**, . . . , **501n** are dialog boxes (e.g., interfaces) of an application (e.g., application **215**), e.g., similar to dialog boxes **311**, **312** of FIGS. 3A-B. Elements **502a**, **502b**, . . . , **502n** are dialogs that contain groups of words included in respective ones of the dialog boxes **501a-n**, e.g., similar to dialogs D1, D2 of FIGS. 3A-B. At step **511**, the glossary relationship set generation module **225** computes feature vectors based on the dialogs **502a-n**. In embodiments, the feature vectors are eigenvectors based on determined TF-IDF (term frequency-inverse document frequency) scores of respective words in the dialogs **502a-n**. At step **512**, the glossary relationship set generation module **225** computes a similarity score of each pair of the feature vectors. In embodiments, the glossary relationship set generation module **225** computes the similarity score using cosine similarity of respective pairs of the feature vectors. At step **513**, the glossary relationship set generation module **225** clusters the dialogs **502a-n** into categories (e.g., C1, C2, C3) using the determined similarities.

(42) With continued reference to FIG. 5, the glossary relationship set generation module **225** generates glossaries **516**, **517**, **517** by extracting words from various documents **519**, **520**, **521**. At step **525**, the glossary relationship set generation module **225** generates a glossary total set by combining the glossaries **516**, **517**, **517** using a union operation. At step **530**, the glossary relationship set generation module **225** performs association analysis on the categories (e.g., C1, C2, C3) to determine glossary relationships **535** for each of the categories. Each one of the categories (e.g., C1, C2, C3) may have one or more glossary relationships. In embodiments, each glossary relationship defines an association of a group of words in the category. In embodiments, the glossary relationship set used at step **402** of FIG. 4 comprises all the glossary relationships of all the categories.

(43) FIG. 6 shows a flowchart of an exemplary method of generating glossary relationships in accordance with aspects of the invention in accordance with aspects of the present invention. Steps of the method may be carried out in the environment of FIG. 2 and are described with reference to elements depicted in FIG. 2-4.

(44) At step **601**, the system generates a base glossary of the application (e.g., application **215**). In embodiments, the glossary relationship set generation module **225** retrieves all resource files

associated with the application and extracts words from the resource files using text extraction. In embodiments, the glossary relationship set generation module **225** filters (e.g., omits) high frequency words (e.g., a, an, the, etc.). The resultant word set is a base glossary of the application, e.g., Glossary.base {w1, w2, w3, . . . }.

(45) At step **602**, the system generates a total glossary for the application. In embodiments, the glossary relationship set generation module **225** combines the base glossary (from step **601**) with one or more other glossaries deemed relevant to this application. Other glossaries may include, for example, an organization glossary (e.g., of an organization that created the application) and a professional glossary (e.g., related to the technical field of the application). In embodiments, the glossary relationship set generation module **225** creates the total glossary using a union function with the plural glossaries. The resultant word set is a total glossary of the application, e.g., Glossary.total {wi}.

(46) At step **603**, the system generates dialogs for the application. In embodiments, the glossary relationship set generation module **225** extracts words from each dialog box of the application (e.g., dialog boxes **311**, **312** of FIGS. 3A-B) and creates groups of the words (e.g., dialogs D1, D2 of FIGS. 3A-B).

(47) At step **604**, the system generates a dialog glossary for each dialog. In embodiments, for each dialog, the glossary relationship set generation module **225** determines a set of words at the intersection of the total glossary and this dialog. The resultant word set is a dialog glossary of the dialog, e.g., Glossary.Di{wi}.

(48) At step **605**, the system performs cluster analysis to determine categories (e.g., classification combinations) of the dialogs. In embodiments, the glossary relationship set generation module **225** determines a feature vector for each respective dialog glossary Glossary.Di{wi}, the feature vector of a particular dialog glossary being a N-dimensional vector, wherein each dimension corresponds to a TF-IDF score of one of the words included in the dialog glossary using Expression 1:

$$(49) \quad TF - IDF(w) = TF(w) * \log\left(\frac{M}{M_w}\right) \quad (1)$$

(50) In Expression 1, TF(w) is the frequency of a word w appearing in this text (e.g., Dialog Di), M is the total number of texts (e.g., Count (Di)), and Mw is the number of texts in which the word w appears (e.g., Count (Di{w})). For a dialog glossary Glossary.Di{wi} containing n words, the feature vector is given by [a1, a2, a3, . . . an], where a1 is the TF-IDF score of the first word, a2 is the TD-IDF score of the first word, etc. Words that do not appear in the dialog glossary Glossary.Di{wi} correspond to a value of zero (0) in the feature vector of this dialog glossary.

(51) Still referring to step **605**, the system determines a cosine similarity between pairs of dialog glossary Glossary.Di{wi} using the determined feature vectors. For example, when the feature vector of a first dialog glossary Glossary.D1 {wi} is given as [x1, x2, x3, . . . , xn] and the feature vector of a second dialog glossary Glossary.D2 {wi} is given as [y1, y2, y3, . . . , yn], then the cosine of an angle between the two is given by Expression 2:

$$(52) \quad \cos = \frac{x_1 y_1 + x_2 y_2 + \text{.Math.} + x_n y_n}{\sqrt{x_1^2 + x_2^2 + \text{.Math.} x_n^2} \cdot \sqrt{y_1^2 + y_2^2 + \text{.Math.} y_n^2}} \quad (2) \quad \text{In cosine similarity, a cosine value of}$$

one (1) is most similar and a cosine value of zero (0) is least similar, with the similarity increasing as the value increases between zero and one. In embodiments, clustering is carried out from the bottom up. In one example, the glossary relationship set generation module **225** calculates the cosine similarity between each pair of dialog glossaries Glossary.Di{wi} and combines the dialog glossaries Glossary.Di{wi} whose cosine similarity is greater than a first (high) threshold value into a sub-category. In this manner, the word group of M is divided into M1 sub-categories, where M1<M. In this example, the module then repeats the process using each determined sub-category. In particular, for each sub-category, the glossary relationship set generation module **225** generates a feature vector for the sub-category (e.g., using a TF-IDF score of each word in the sub-category). The glossary relationship set generation module **225** then determines cosine similarity of pairs of

the sub-categories using the determined feature vectors of the sub-categories. Using the determined cosine similarities, the glossary relationship set generation module **225** combines ones of the sub-categories that are sufficiently similar (e.g., their cosine similarity score exceeds the high threshold) into larger categories, where the number of larger categories is less than the number of sub-categories. The glossary relationship set generation module **225** iterates this process, thus creating fewer categories, until the categories are not sufficiently similar to combine (e.g., their similarity score is less than a second (low) threshold). Finally, all dialog words are classified and category glossaries are generated. The result is a group of categories, where each category contains a group of dialogs as shown in Expression 3, and a category glossary for each category as shown in Expression 4:

Category.sub.c1,{D1,D5,D6, . . . },Category.sub.c2,{D2,D3,D4, . . . },Category.sub.c3{D7,Dg, . . . } (3)

Glossary.sub.c1,{w1,w2,w3, . . . },Glossary.sub.c2,{w1,w4, . . . },Glossary.sub.c3{w7,w8, . . . } (4)

(53) In the examples shown in Expression 3 and Expression 4, the words {w1, w2, w3, . . . } in category glossary GlossaryC1 are all the words in the dialogs {D1, D5, D6, . . . } of CategoryC1. Similarly, GlossaryC2 is a set of all the words in the dialogs {D2, D3, D4, . . . } of CategoryC2. In this manner, the glossary relationship set generation module **225** uses clustering and to generate plural categories (e.g., CategoryC1, CategoryC2, etc.) and a category glossary for each category (e.g., GlossaryC1, GlossaryC2, etc.). In embodiments, the different categories (e.g., CategoryC1, CategoryC2, etc.) are mutually exclusive meaning that each dialog (e.g., D1, D2, etc.) appears in only one of the categories.

(54) At step **606**, the system uses association analysis using the determined categories (e.g., CategoryC1, CategoryC2, etc.) and category glossaries (e.g., GlossaryC1, GlossaryC2, etc.) to determine one or more glossary relationships for each category. In embodiments, the glossary relationship set generation module **225** uses market basket analysis techniques to determine groups of words in a category glossary that have a support index greater than a support threshold. In embodiments, the support threshold is 0.3 and the support index is shown by Expression 5:

(55) $\text{Support}\{w_x, w_y, w_z, \} = \frac{\text{Count}(D\{w_x, w_y, w_z, \})}{\text{Count}(\text{Category}_t\{D\})} > 0.3$ (5)

(56) Using this technique, the glossary relationship set generation module **225** determines groups of words that are associated with one another, these groups referred to glossary relationships Glossary.Ci.Rn {wx, wy, wz}. In embodiments, the glossary relationship set generation module **225** determines plural glossary relationships for each category (e.g., CategoryC1, CategoryC2, etc.). In embodiments, the glossary relationship set used at step **402** of FIG. 4 is all the glossary relationships for all categories.

(57) FIG. 7 shows an example of determining the glossary relationships for CategoryC1. In this example, the glossary relationship set generation module **225** determines category CategoryC1 (shown at **701**) includes dialogs D1, D5, and D6, e.g., using clustering as described herein. In this example, the glossary relationship set generation module **225** determines GlossaryC1 (shown at **702**) based on the words included in CategoryC1, and then determines glossary relationships (shown at **703**) for this category using association analysis as described herein. In this example, one of the determined glossary relationships is the group of words {OK, Paste, Reset, Cancel, Help} in dialog D1. For reference, this grouping of words is shown in dialog box **311** that corresponds to dialog D1.

(58) In embodiments, the glossary relationship set generation module **225** maps two or more dialogs to the same glossary relationship in response to the glossary relationship set generation module **225** determining the same glossary relationship for each of the two or more dialogs. For example, the glossary relationship set generation module **225** determines glossary relationship1 {OK, Paste, Reset, Cancel, Help} in dialog D1 of dialog box **311** (of FIG. 3A) using the clustering

and association analysis techniques as described herein. Because dialog box **312** of FIG. **3B** includes the same group of words (i.e., OK, Paste, Reset, Cancel, Help), the glossary relationship set generation module **225** will arrive at the same glossary relationship **1** {OK, Paste, Reset, Cancel, Help} in dialog **D2** of dialog box **312**. Based on these same determined glossary relationships, the glossary relationship set generation module **225** determines a mapping of the different dialogs to the same glossary relationship.

(59) FIG. **8** shows an exemplary mapping **800** of dialogs to glossary relationships in accordance with aspects of the invention. As shown in FIG. **8**, dialog **D5** included in Category **C1** is mapped to glossary relationship **1**. Also in this example, dialog **D2** included in Category **C2** is mapped to glossary relationship **1**. In embodiments, the system uses this mapping to determine which instances of a word to change the translation of in response to receiving user input as described herein.

(60) FIG. **9** shows a flowchart of an exemplary method in accordance with aspects of the present invention. Steps of the method may be carried out in the environment of FIG. **2** and are described with reference to elements depicted in FIG. **2-8**.

(61) At step **905**, the system generates a glossary relationship set for an application. In embodiments, the glossary relationship set generation module **225** generates a dialog for each interface (e.g., dialog box) of the application, wherein the dialog for a respective interface is a list of words appearing in the respective interface. In embodiments, the glossary relationship set generation module **225** groups the dialogs into categories using clustering. In one example, the clustering is based on cosine similarity. In embodiments, the glossary relationship set generation module **225** generating a respective glossary for each of the categories. In embodiments, for each category, the glossary relationship set generation module **225** determines glossary relationships for the category by performing association analysis using the glossary of the category and the dialogs included in the category. In one example, the association analysis is a market basket analysis.

(62) At step **910**, the system receives user input indicating an incorrect translation of a word appearing in an interface of the application. In embodiments, a user that is utilizing application **215** on their user device **205** provides input (e.g., via a translation feedback interface shown in the user interface **223**) that a word appearing in one of the interfaces (e.g., dialog box **311**) is incorrectly translated. In embodiments, the translation server **210** receives this user input via the network **201**. The user input may include the incorrectly translated word and a correct translation of the word.

(63) At step **915**, the system identifies other instances of the word in other interfaces of the application using a glossary relationship set that is based on association analysis. In embodiments, the relationship application module **230** uses the glossary relationship set for this application **215** to identify other instances of the same word in other interfaces of the application **215**. In embodiments, the relationship application module **230** identifies a glossary relationship that is associated with a dialog that contains the word indicated by the user, and uses a mapping (e.g., as shown in FIG. **8**) to determine every other dialog that is mapped to the same glossary relationship and that has an instance of the word in the dialog. A dialog that has an instance of the word but that is not mapped to the same glossary relationship is ignored at this step, such that the instance of the word in that dialog will not have its translation changed.

(64) At step **920**, the system generates a new version of the application having a revised translation of the word in the interface and the other instances of the word in the other interfaces. In embodiments, the translation engine module **235** revises a translation of the word and the other instances of the word in portions of a resource file that defines the interface and the other interfaces. In embodiments, the modification and compiler module **240** generates a new version of the application **215** having a revised translation of the word in the interface and the other instances of the word in the other interfaces. In embodiments, the verification module **245** tests the new version of the application **215** using a translation verification test. The translation server **210** may deliver the new version of the application **215** to the user device **205**.

(65) In embodiments, a service provider could offer to perform the processes described herein. In

this case, the service provider can create, maintain, deploy, support, etc., the computer infrastructure that performs the process steps of the invention for one or more customers. These customers may be, for example, any business that uses technology. In return, the service provider can receive payment from the customer(s) under a subscription and/or fee agreement and/or the service provider can receive payment from the sale of advertising content to one or more third parties.

(66) In still additional embodiments, the invention provides a computer-implemented method, via a network. In this case, a computer infrastructure, such as computer system **12** (FIG. **1**), can be provided and one or more systems for performing the processes of the invention can be obtained (e.g., created, purchased, used, modified, etc.) and deployed to the computer infrastructure. To this extent, the deployment of a system can comprise one or more of: (1) installing program code on a computing device, such as computer system **12** (as shown in FIG. **1**), from a computer-readable medium; (2) adding one or more computing devices to the computer infrastructure; and (3) incorporating and/or modifying one or more existing systems of the computer infrastructure to enable the computer infrastructure to perform the processes of the invention.

(67) The descriptions of the various embodiments of the present invention have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

Claims

1. A method, comprising: receiving, by a computing device, user input indicating an incorrect translation of a word appearing in a dialog box of an application; identifying, by the computing device, other instances of the word in other dialog boxes of the application, wherein the identifying is performed using a glossary relationship set that is based on association analysis, wherein the other instances of the word in the other dialog boxes constitute less than all instances of the word appearing in all dialog boxes of the application; and generating, by the computing device, a new version of the application having a revised translation of the word in the dialog box and the other instances of the word in the other dialog boxes, wherein the word appearing in the dialog box, and the other instances of the word in the other dialog boxes, are in a first language and are translated from a resource file of the application written in second language different than the first language.
2. The method of claim 1, wherein the identifying comprises determining that respective groups of words containing the other instances of the word are mapped to a same glossary relationship in the glossary relationship set.
3. The method of claim 1, wherein the generating the new version of the application comprises: revising a translation of the word and the other instances of the word in portions of the resource file that defines the dialog box and the other dialog boxes; and compiling, assembling, and integrating the new version of the application after the revising.
4. The method of claim 1, further comprising testing the new version of the application using a translation verification test.
5. The method of claim 1, further comprising generating the glossary relationship set.
6. The method of claim 5, wherein the generating the glossary relationship set comprises: generating a dialog for each dialog box of the application, wherein the dialog for a respective dialog box is a list of words appearing in the respective dialog box; grouping the dialogs into categories using clustering; generating a respective glossary for each of the categories; and for each one of the categories, determining glossary relationships for the one of the categories by

performing association analysis using the glossary of the one of the categories and the dialogs included in the one of the categories.

7. The method of claim 6, wherein the clustering uses cosine similarity.

8. The method of claim 6, wherein the association analysis uses market basket analysis.

9. A computer program product comprising one or more computer readable storage media having program instructions collectively stored on the one or more computer readable storage media, the program instructions executable to: receive user input indicating an incorrect translation of a word appearing in an interface of an application; identify other instances of the word in other interfaces of the application, wherein the identifying is performed using a glossary relationship set that is based on association analysis, wherein the other instances of the word in the other interfaces constitute less than all instances of the word appearing in all interfaces of the application; and generate a new version of the application having a revised translation of the word in the interface and the other instances of the word in the other interfaces, wherein the interface comprises one of plural dialog boxes of the application; the other interfaces comprise other ones of the plural dialog boxes of the application; and the word appearing in the interface, and the other instances of the word in the other interfaces, are in a first language and are translated from a resource file of the application written in second language different than the first language.

10. The computer program product of claim 9, wherein the identifying comprises determining that respective groups of words containing the other instances of the word are mapped to a same glossary relationship in the glossary relationship set.

11. The computer program product of claim 9, wherein the generating the new version of the application comprises: revising a translation of the word and the other instances of the word in portions of the resource file that defines the interface and the other interfaces; and compiling, assembling, and integrating the new version of the application after the revising.

12. The computer program product of claim 9, wherein the program instructions are executable to test the new version of the application using a translation verification test.

13. The computer program product of claim 9, wherein: the glossary relationship set comprises a set of plural glossary relationships where each respective one of the plural glossary relationships defines a group of associated words from a respective one of the plural dialog boxes of the application.

14. The computer program product of claim 13, wherein the generating the glossary relationship set comprises: generating a dialog for each interface of the application, wherein the dialog for a respective interface is a list of words appearing in the respective interface; grouping the dialogs into categories using clustering based on cosine similarity; generating a respective glossary for each of the categories; and for each one of the categories, determining glossary relationships for the one of the categories by performing association analysis using the glossary of the one of the categories and the dialogs included in the one of the categories.

15. A system comprising: a processor, a computer readable memory, one or more computer readable storage media, and program instructions collectively stored on the one or more computer readable storage media, the program instructions executable to: receive user input indicating an incorrect translation of a word appearing in an interface of an application; identify other instances of the word in other interfaces of the application, wherein the identifying is performed using a glossary relationship set that is based on association analysis, wherein the other instances of the word in the other interfaces constitute less than all instances of the word appearing in all interfaces of the application; and generate a new version of the application having a revised translation of the word in the interface and the other instances of the word in the other interfaces, wherein the new version of the application does not change all instances of the word appearing in all interfaces of the application to the revised translation of the word, wherein the interface comprises one of plural dialog boxes of the application; the other interfaces comprise other ones of the plural dialog boxes of the application; and each of the plural dialog boxes is associated with a respective different

function of the application.

16. The system of claim 15, wherein the identifying comprises determining that respective groups of words containing the other instances of the word are mapped to a same glossary relationship in the glossary relationship set.

17. The system of claim 15, wherein the generating the new version of the application comprises: revising a translation of the word and the other instances of the word in portions of a resource file that define the interface and the other interfaces; and compiling, assembling, and integrating the new version of the application after the revising.

18. The system of claim 15, wherein the program instructions are executable to test the new version of the application using a translation verification test.

19. The system of claim 15, wherein the program instructions are executable to generate the glossary relationship set, and wherein the user input comprises the word, a revised translation of the word, and an indication of a dialog box in which the word appears.

20. The system of claim 19, wherein the generating the glossary relationship set comprises: generating a dialog for each interface of the application, wherein the dialog for a respective interface is a list of words appearing in the respective interface; grouping the dialogs into categories using clustering based on cosine similarity; generating a respective glossary for each of the categories; and for each one of the categories, determining glossary relationships for the one of the categories by performing association analysis using the glossary of the one of the categories and the dialogs included in the one of the categories.
